

UNIT 3: PRACTICE QUESTION BANK & ANSWER KEY

Course Pattern: TE IT - 2024 Pattern (SPPU)
Document Type: Question Bank with Answer Key

Academic Term: Semester V (2026)
Course Code: PCC-301-ITT

Section A: Important Theory Questions

- Critical Section Problem & Synchronization:**
 - Define a Race Condition with an example. Explain the three formal criteria required to solve the Critical Section Problem. [SPPU Nov 2023, Marks: 8]
 - Explain the hardware-assisted atomic instructions TestAndSet and CompareAndSwap. How do they enforce mutual exclusion? [SPPU May 2024, Marks: 6]
- Semaphores and Classical Synchronization:**
 - Differentiate between Counting Semaphores and Binary Semaphores. Explain the non-busy-waiting implementation of wait() and signal() operations. [SPPU Dec 2022, Marks: 8]
 - Provide a complete semaphore-based solution for the Bounded-Buffer Producer-Consumer Problem. [SPPU Nov 2023, Marks: 8]
 - Describe the Dining Philosophers Problem. Explain how deadlock occurs and how it can be prevented using semaphores. [SPPU May 2023, Marks: 8]
- Deadlocks and Handling Strategies:**
 - State and explain the four Coffman conditions necessary for a deadlock to occur. [SPPU May 2024, Marks: 6]
 - Differentiate between Deadlock Prevention, Deadlock Avoidance, and Deadlock Detection. [SPPU Dec 2023, Marks: 8]

Section B: Numerical Practice Problems

Practice Problem 1: Banker's Algorithm Safety Test

Consider a system with 5 processes (P_0, P_1, P_2, P_3, P_4) and 3 resource types (A, B, C). Total instances: $A = 10, B = 5, C = 7$.

At time t_0 :

Process	Allocation (A B C)	Max (A B C)	Available (A B C)
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

- (a) Compute the Need Matrix. [Marks: 4]
- (b) Find the Safe Sequence using the Safety Algorithm. [Marks: 4]

Practice Problem 2: Resource Request Algorithm

Using the system snapshot from Practice Problem 1:

- If process P_1 requests $(1, 0, 2)$, can the request be granted immediately? Show all intermediate calculations. [SPPU Nov 2023, Marks: 8]

Mandatory Final Answer Key

Below are the verified solutions for self-verification:

- **Practice Problem 1 (a) — Need Matrix:**
 - $\text{Need}[P_0] = (7, 4, 3)$
 - $\text{Need}[P_1] = (1, 2, 2)$
 - $\text{Need}[P_2] = (6, 0, 0)$
 - $\text{Need}[P_3] = (0, 1, 1)$
 - $\text{Need}[P_4] = (4, 3, 1)$
- **Practice Problem 1 (b) — Safety Algorithm:**
 - Initial Work = $(3, 3, 2)$.
 - P_1 executes to Work = $(5, 3, 2)$.
 - P_3 executes to Work = $(7, 4, 3)$.
 - P_4 executes to Work = $(7, 4, 5)$.
 - P_0 executes to Work = $(7, 5, 5)$.
 - P_2 executes to Work = $(10, 5, 7)$.
 - **System Status:** Safe State.
 - **Safe Sequence:** $\angle P_1, P_3, P_4, P_0, P_2 r \angle$.
- **Practice Problem 2 — Resource Request:**
 - $\text{Request}[P_1](1, 0, 2) \leq \text{Need}[P_1](1, 2, 2)$ (True) and $\leq$ Available $(3, 3, 2)$ (True).
 - Tentative State: Available = $(2, 3, 0)$, Allocation[P_1] = $(3, 0, 2)$, Need[P_1] = $(0, 2, 0)$.
 - Safety check succeeds with Safe Sequence $\angle P_1, P_3, P_4, P_0, P_2 r \angle$.
 - **Decision:** Request Granted immediately.

End of Operating Systems Unit 3 Practice Question Bank & Answer Key — SPPU TE IT 2024 Pattern